SQL Functions for Data Cleaning

Clean and Optimize Your Data Using SQL



Why Data Cleaning is Important?

- • Ensures accuracy and consistency in your data.
- • Prepares data for analysis and reporting.
- • Eliminates errors, duplicates, and inconsistencies.

Data Issues that Need Cleaning

- • Duplicates
- • Inconsistent Formats
- • Null Values
- • Incorrect Data Types
- • Outliers
- • Extra Whitespaces

Overview of Useful SQL Functions

- • TRIM: Removes leading and trailing spaces.
- • REPLACE: Substitutes part of a string.
- • COALESCE: Replaces null values.
- • CAST/CONVERT: Changes data types.
- • SUBSTRING: Extracts part of a string.
- • UPPER/LOWER: Standardizes text to uppercase or lowercase.

Removing Duplicate Rows

- • Function: SELECT DISTINCT
- Example:

SELECT DISTINCT column_name

FROM table_name;

Use Case: Ensures each record in a result set is unique.

Replacing Null Values

- • Function: COALESCE
- Example:

```
SELECT COALESCE(column_name, 'default_value')
FROM table_name;
```

- • Use Case: Replace nulls with default values for clean reports.

Consistent Data Formatting

- • Functions: UPPER, LOWER, CAST
- • Example:

SELECT UPPER(column_name), CAST(column_name AS INT) FROM table_name;

- • Use Case: Ensure consistent format across your dataset.

Cleaning Extra Whitespaces

- • Function: TRIM
- • Example:

```
SELECT TRIM(column_name)
FROM table_name;
```

- • Use Case: Remove unwanted spaces from text data.

Replacing Incorrect Text

- • Function: REPLACE
- Example:

```
SELECT
REPLACE(column_name,
'old_text', 'new_text')
FROM table_name;
```



- • Use Case: Correct misspellings or standardize values.

Summary of SQL Cleaning Functions



- • TRIM: Removes spaces.
- • REPLACE: Substitutes strings.
- • COALESCE: Handles nulls.
- • DISTINCT: Removes duplicates.
- • CAST: Changes data types.



- Stay connected for more SQL tips and data cleaning techniques!